Jan Gałek

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Education

University of Oxford (Oxford, UK) 2025–2026
MSc Mathematics and Foundations of Computer Science **Distinction**

Selected marks: Applied Category Theory (93/100); Categories, Proofs and Processes (92/100); Lambda Calculus and Types (82/100); Computer-aided Formal Verification (78/100).

University of Warwick (Coventry, UK) 2022–2025
BSc Mathematics and Philosophy **First Class Honours**

Research Experience

MSc Dissertation — University of Oxford 2026 | 84/100
Contextual Equivalence for Simply Typed Lambda Calculi *Dissertation PDF linked*

- Constructed a finite, effectively presentable denotational model for simply typed λ-calculi with finite ground signatures of at least two constants; established **full abstraction** and derived a **semantic decision procedure for contextual equivalence**.
- Introduced **finite-component retractions and retraction bases**; proved transfer principles reducing higher-order problems concerning decidability, universality and full abstraction to simpler basis types.
- Applied the framework to unary PCF, unary PCF with parallel composition and Boolean PCF; used Loader's undecidability theorem to demonstrate an obstruction to effective low-order compression.

Technical Background

Programming-language theory: Denotational semantics, type systems, λ-calculus, contextual equivalence, categorical methods.

Automated verification coursework: Temporal logic, model checking, symbolic methods and SAT-based verification (Oxford CAFV course).

Programming: Python (academic coursework with NumPy, pandas, scikit-learn and Keras); individual machine-learning assignment (92/100).

Technical writing: LaTeX.

Additional Experience and Honours

Self-employed Mathematics Tutor 2022–present
Explained mathematical concepts and developed students' problem-solving skills through individual tutoring.

Honours: Finalist, Polish National Philosophy Olympiad (2019, 2020).

Languages: Polish (native), English (advanced), French (basic).

Academic references available upon request.